

New Phytologist

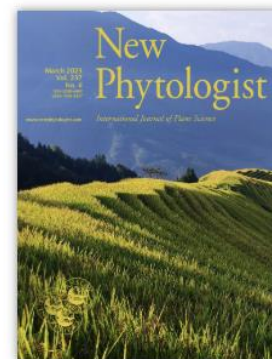
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Why can't we predict traits from the environment?

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Leander Anderegg is the joint winner, with Moi Exposito-Alonso, of the 2022 *New Phytologist* Tansley Medal for excellence in plant science. See Slater & Dolan in this issue of *New Phytologist* (**2023**, **237**: 1939–1940) for more details.



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Figure 1: Environment is a poor predictor of traits

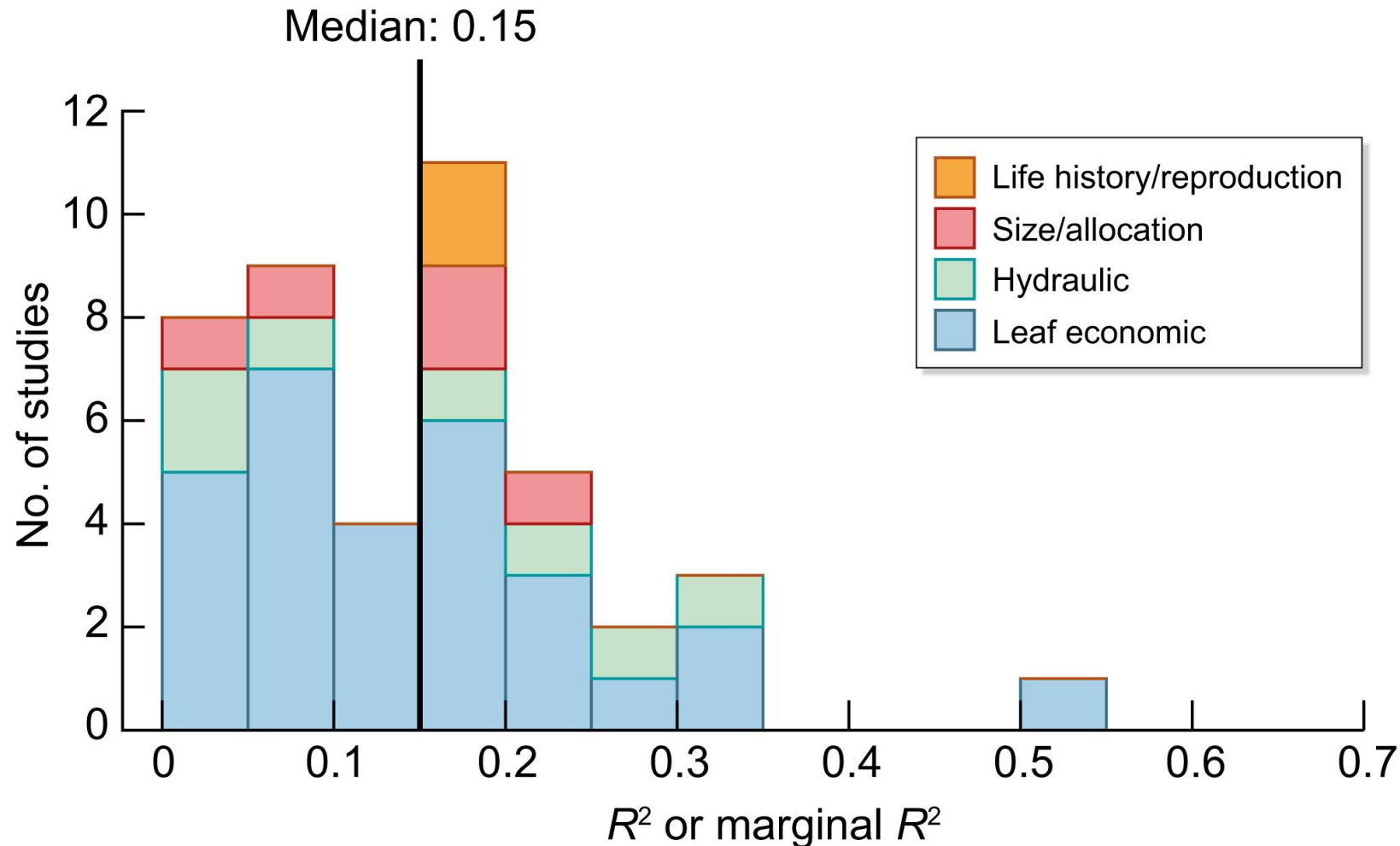


Figure 2: Scales complicate trait-environment predictions

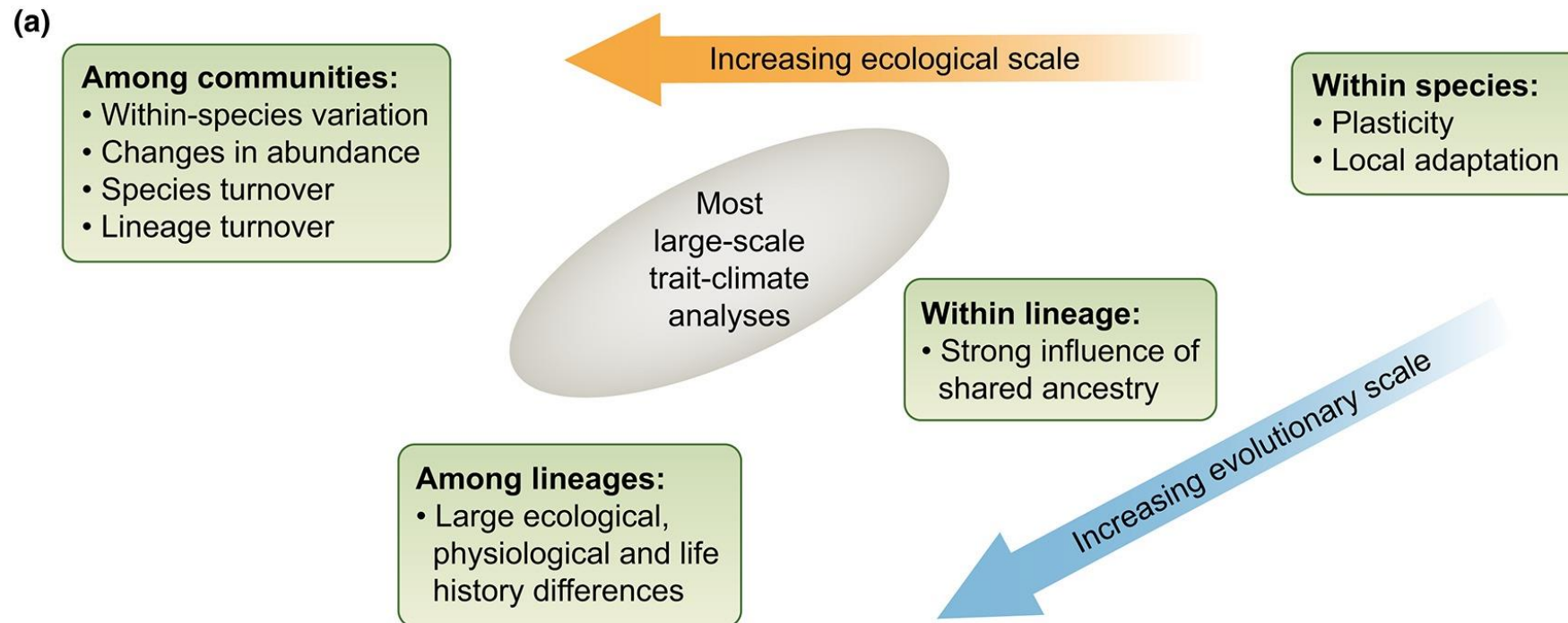


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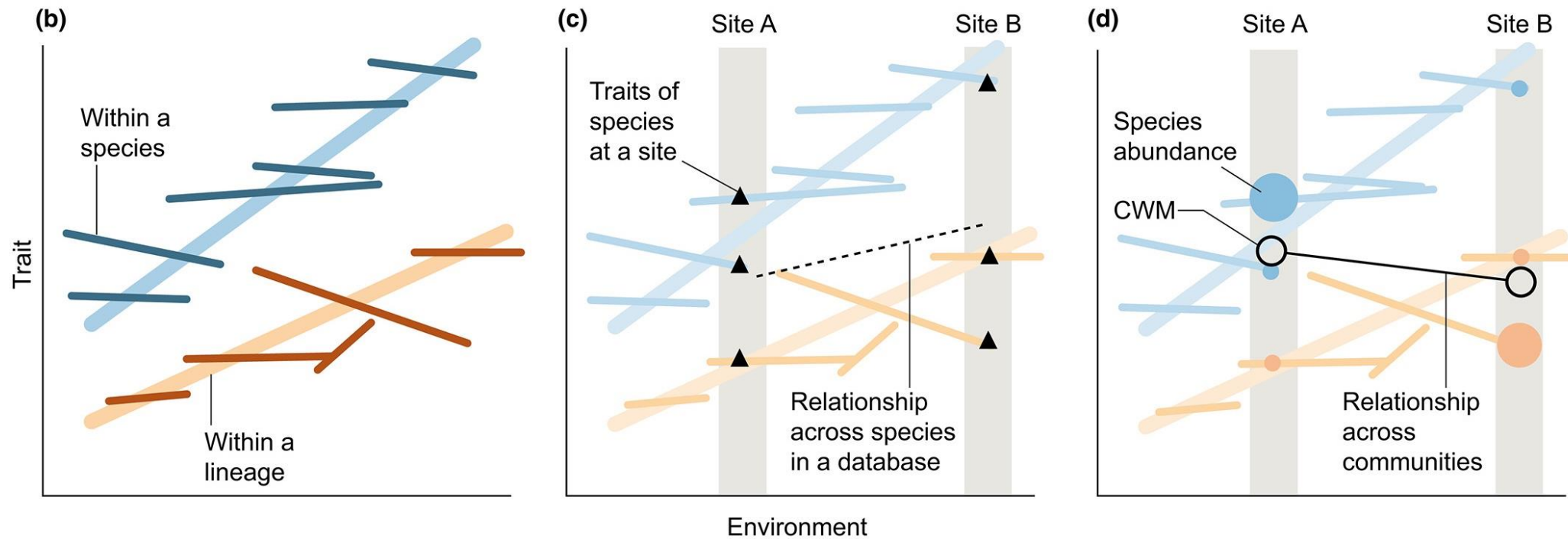


Table 1

	Hypotheses for weak trait–environment relationships	Potential workarounds
Within species	Nonadaptive trait variation, such as ‘negative plasticity’ (a stress-induced change in phenotype that decreases fitness) and genetic drift among populations or founder-effects in small range-edge populations (Fig. 2b) Microclimatic or micro-edaphic variation decouples the environment that sampled individuals actually experience from environmental predictors used in analysis Incomplete sampling of species’ environmental niche hides true trait patterns Confounding stress gradients have opposing effects on focal traits but are difficult to disentangle geographically	• Common garden studies to disentangle plasticity and ecotypic variation • Sample more populations to overcome noise • Careful site selection using multiple climate variables
Among species	‘Phylogenetic noise’ of sampling across lineages obscures consistent patterns within lineages (Fig. 2c) Measures of species climate niche are much stronger predictors of relative trait values among species than the site-of-observation climate for any particular trait measurement Equal influence of common species and rare species with outlier trait values obscures trait signal	• Phylogenetically informed analyses • Use metrics of species’ climate from occurrence data as predictors • Calculate CWM traits to account for abundance
Among communities	Lineage turnover among communities complicates the functional significance of an absolute change in trait value (Fig. 2d) Conflicting and countergradient patterns at multiple ecological scales sum to inconclusive community patterns	• Hierarchical analyses to decompose effects of lineage turnover, species turnover, and within-species variation
All levels	Measured traits are ‘functional’ (i.e. they affect fitness) along orthogonal ecological axes to simple climate or soil gradients, and thus only indirectly relate to abiotic environmental variables Unmeasured/underappreciated traits are more important or ‘functional’ than the most easily measured traits Trait integration/equifinality among many trait combinations minimizes univariate relationships with any individual trait even while functional phenotype changes strongly with environment	• Explore the whole-organism or integrated phenotype, including the effects of many traits (e.g. by using mechanistic models to understand trait equifinality and performance landscapes)

Troublesome traits

- “we poorly understand the **functional significance** of traits themselves”
- “our understanding of **how plant traits...influence...performance**...is still evolving. We may be focusing on the wrong traits”
- “There are **many successful ways to be a plant** in most environments”

Path forward

- Hierarchical analysis needed for mechanisms (scale matters)
- “Function” of functional traits needs to be better understood
- Phenotypes are integrated
 - Equifinality
 - Models needed to generate hypotheses

